

Operations Research 2016/2017

Date: 26/01/2017

Exam – 1st Call

Duration: 2 hours

Note: Present **all the calculations** that you perform and appropriately **justify** your answers.

1. Consider the following problem:

“The company SAVENERGY of Coimbra has a project that intends to estimate the extent of the energy loss in homes and companies of the district. Instead of setting the number of homes and companies to inspect, the company has assigned to this project three experts who will inspect the insulation, electrical installation, and heating system of a set of residences and companies for the duration of the project.

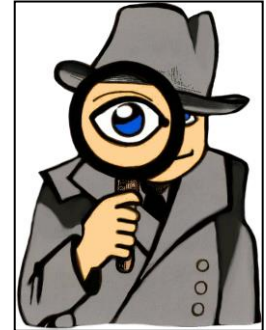
The company considers that in order for the obtained information to be useful, a complete inspection should include the three aforementioned aspects (insulation, electrical installation and heating system).

The insulation inspector estimates that it will take 4 hours to fully inspect a home and only 2 hours to inspect a company. In turn, the electrical inspector estimates that it will take 2 hours to inspect a home and 6 hours to inspect a company. Finally, the heating inspector estimates that it will take 4 hours to inspect a home and 6 hours to inspect a company.

On the other hand, the company expects that the weekly availability for inspection work will be 28 hours in the case of the insulation inspector, 30 hours in the case of the electrical inspector and 36 hours in the case of the heating inspector.

The company has set as goal to maximize the number of completed inspections per week.”

To help SAVENERGY, formulate the problem in terms of a linear programming model, indicating the meaning of the decision variables and the objective function..



2. Consider the following linear programming problem:

$$\text{Minimize } z = 4x_1 + 3x_2$$

subject to

$$-x_1 + 2x_2 \leq 4$$

$$2x_1 - 2x_2 \geq 6$$

$$2x_1 + 2x_2 = 10$$

$$x_1 \geq 0, x_2 \geq 0$$

- Solve it by the graphical method;
- Solve it again by the Simplex method using the “Two Phases” technique and identify, in each iteration, the corresponding basic solution;
- Say when two feasible basic solutions are adjacent and give an example taken from the previous resolution in b).

Scores: 1 – 3,5 points 2 – 5,5 points 3 – 5,5 points 4 – 5,5 points

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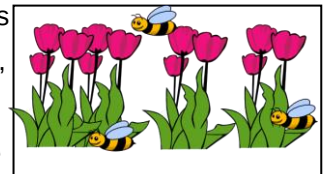
3. Consider now the following linear programming problem:

Maximizar $z = x_1$
 sujeito a
 $2x_1 + x_2 \leq 2$
 $x_1 + 3x_2 \leq 3$
 $x_1 \geq 0, x_2 \geq 0$

- Solve it by the dual Simplex method using the “Big M” technique;
- Formulate the dual problem corresponding to the problem presented above;
- Explain the difference of the graphical operation of Simplex and dual Simplex methods in terms of the sequence of extreme points (EPs) followed when solving any linear programming problem.

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4. A given company that imports tulips from Holland has three warehouses (designated **W1**, **W2** and **W3**), which temporarily house the imported flowers, and three stores in the center of the country (designated **S1**, **S2**, and **S3**), where they are placed for sale. It is known that **W1**, **W2** and **W3** dispose weekly of **21**, **18** and **11** boxes of tulips, respectively. On the other hand, it is known that the weekly requirements by stores **L1**, **L2** and **L3** are of **15**, **20** and **10** boxes of tulips, respectively. The costs of transporting each box of tulips between the various warehouses and the various stores of the company are also known. These costs (in euros, per box) are shown in the following table:



	S1	S2	S3
W1	1	2	5
W2	8	3	1
W3	4	9	6

- Obtain an initial basic feasible solution for the problem by the Northwest Corner method;
- Starting from the solution obtained in a) solve the problem by the transportation method;
- Indicate the constraints of the primal problem corresponding to the variables u_i of the dual problem used in the resolution of b).